Accepted Manuscript
Version of Record at: https://doi.org/10.1021/acs.analchem.3c04404

Total and class-specific determination of fluorinated compounds in consumer and food packaging samples using fluorine-19 solid state nuclear magnetic resonance spectroscopy

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ABSTRACT: Hamburger wrapping paper, coated with water-based barrier coatings, used in the food packaging industry was studied by using the total organic fluorine (TOF) method, based on combustion ion chromatography (CIC), and fluorine-19 solid-state nuclear magnetic resonance (^{19}F ss-NMR) spectroscopy. Although the TOF method is a fast and affordable method used to screen for per- and polyfluoroalkyl substances (PFAS), the amount of fluorine it measures is heavily dependent on the extraction step and, therefore could lead to inaccurate results. Fluorine-19 ss-NMR spectroscopy can differentiate between organic and inorganic fluorinated sources, eliminating the need for sample clean up. To illustrate this, the ^{19}F ss-NMR spectra of clean coated paper samples that contained naturally occurring F- ions from the talc raw material and spiked samples containing perfluorooctanoic acid (PFOA) were compared. A range of experimental conditions was explored to improve sensitivity for low PFAS concentrations (in the order of 10 – 20 mg/kg). Despite the disadvantages of ss-NMR spectroscopy, such as low limit of detection (LOD) and resolution, the results demonstrate that it is a viable tool to directly detect PFAS moieties in consumer and food packaging. Therefore, 19-F solid-state NMR spectroscopy challenges and complements current methods, which only provide indirect evidence of the presence of PFAS.

INTRODUCTION

Per- and Polyfluoroalkyl Substances

Organofluorines are organic compounds that contain at least one carbon-fluorine bond. These bonds are mostly found in anthropogenic compounds such as pharmaceuticals, agrochemicals, refrigerants, polymers, surfactants, and so forth. Per- and polyfluoroalkyl substances (PFAS) are a subclass of organofluorines that contain a perfluoroalkyl moiety (C_nF_{2n+1}-) with at least three carbons, although multiple PFAS definitions exist.[1–4] Carbon–fluorine bonds are one of the strongest bonds in organic chemistry because of their partial ionic character due to the high electronegativity of fluorine, causing high dipole moments.[5] When more fluorine atoms are bonded to the same carbon atom, the C–F bonds become stronger and shorter. These stronger, shorter bonds make the perfluoroalkyl moieties in PFAS chemically inert.[6] Being less reactive makes them extremely useful in a wide range of commercial, industrial and consumer products because the compounds are chemically inert, have a high thermal stability and are simultaneously hydro- and lipophobic. Their numerous applications, multiple emission routes and resistance to degradation processes lead to a remarkable persistence in the environment, a multitude of exposure pathways and the potential for adverse effects on wildlife and human health, hence earning them the nickname "forever chemicals".[7,8] Although most PFAS exhibit low acute toxicity, their gradual accumulation due to continuous exposure can lead to a wide range of health effects in which they can affect the immune, digestive, metabolic, endocrine, reproductive, and nervous systems.[9–11] Therefore, perfluorooctanesulfonic acid (PFOS), perfluorooctanoic acid (PFOA) and their related substances for example were the first PFAS to be regulated by the Stockholm Convention on Persistent Organic Pollutants (POPs; Decision SC-4/17; Decision SC-9/12), the E.U. European Chemicals Agency (ECHA; EU 207/2011, EU 2017/1000, EU 2019/1021) and the U.S. Environmental Protection Agency (EPA; FR 2016-12361, FR 2022-13158).

Regulation, however, forced producers to move to other types of PFAS, which led to an increase in the variation of their structural compositions. According to the OECD, around 5000 distinct PFAS have been identified and categorized by CAS Registry Number; however, the EPA's toxicity database, DSSTox, lists around 15000 different compounds.[2,12] This discrepancy indicates that a myriad of unidentified and therefore less-studied PFAS exist, each with its own impurities, intermediates and precursors.[13,14]

PFAS Applications and Life Cycle

Many different applications, including firefighting foams, outdoor clothing, cosmetics, and cookware use PFAS and fluorosurfactants to lower the surface tension to provide film forming properties and water-, grease- or stain-resistance.[15,16] These chemicals are also widely used in the pulp and paper industry to produce excellent water repelling, grease resistance and release properties in food packaging such as take-away containers, fast-food wrappers, baking papers, and microwavable popcorn bags.[17,18] People are exposed directly to PFAS when the chemicals, because of their solubility in water and grease, migrate from consumer goods and food packaging

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into the foodstuff.[19,20] The high turnover rates of consumer goods and food packaging mean that the PFAS quickly end up in landfills and in waste infrastructures.[21,22] Therefore, these chemicals can enter the environment through leachates and effluents and pollute our food, drinking (surface and ground) water and, indirectly, our bodies.[23–26] Thus, PFAS in fiber-based substrates are concerning because people are exposed to emissions at every stage of the PFAS life cycle.

Food Contact Materials and PFAS Legislation

Although clear evidence exists that the use of PFAS in food contact materials (FCMs) is problematic, a harmonized legislation does not yet exist. However, Denmark, Germany, the Netherlands, Norway, and Sweden submitted a restriction proposal with ECHA in February 2023 aimed at reducing or banning emissions of PFAS from all sources, especially in FCMs.[27,28] The existing framework for FCMs and packaging papers, including the E.U. regulation (EC) No 1935/2004, the U.S. Food and Drug Administration's (FDA) Title 21 Code of Federal Regulations (CFR) 175.170 and CFR 175.180 and Germany's BfR XXXVI recommendations, only provide general safety requirements and recommendations. Therefore, safety measures for specific substances, such as PFAS, only exist for very few of these packaging materials or different rules apply across different countries, regions or states.[18,29] For example, since July 2020 Denmark has a regulation in place that specifically bans PFAS in food contact paper and board packaging for both direct and indirect uses. To implement this legislation, the Danish authorities developed a test method called the total organic fluorine (TOF) method. To differentiate between intentionally added PFAS (both direct and indirect) and environmental background levels, a guided indicator value was fixed at 20 mg/kg dry weight (dw) TOF.[17,30]

The Total Organic Fluorine Method

The TOF method is based on combustion ion chromatography (CIC), wherein a sample undergoes pyrolysis to convert all fluorinated substances into hydrogen fluoride (HF). The HF is then collected and determined by measuring the conductivity. Because this technique only provides total fluoride (TF) concentrations and does not provide any structural information, the method is non-specific for all fluorinated compounds, both organic and inorganic.[31–34] Therefore, the organic fluorine (OF) fraction must be determined by subtracting the inorganic fluoride (IF) fraction from the total fluorine (TF – IF = OF). The IF fraction in turn, is determined by IC of a cold-water liquid extract. Since this method measures all OF in a sample, it is considered to be an acceptable proxy for the total concentration of PFAS in fiber-based substrates because other organofluorines are rarely used in food and consumer packaging.[17] The biggest advantages of the TOF method are its simplicity, low limit of quantification (LOQ = 0.01 μg F^- absolute for CIC) and relatively low cost, which make it quite suitable for test service laboratories to test for compliance.[32,35,36] The drawbacks for this method are inaccurate results due to interference of chloride ions, fouling of alkali elements, and the low precision in the determination of OF when the IF fraction is much higher than the OF fraction.[37,38] Also, the presence of non-extractable fluorines in a sample will be interpreted as TOF (vide infra). These drawbacks can cause underestimation or overestimation of the TOF fraction. This could lead to consumer and food packaging materials that still contain PFAS being introduced into the market or banning the articles by the upcoming ECHA legislation that are actually PFAS-free and therefore perfect alternatives for current solutions.[17]

Total Analysis of Fluorinated Substances

In general, analysis of fluorinated substances in fiber-based substrates are not straight-forward. No single method is versatile, robust, common, and sensitive enough to quantify and identify the whole group of PFAS, their impurities and degradation products, and to differentiate from other fluorine containing chemicals (fluorides, agrochemicals and pharmaceuticals) that could be present in a sample.[33,34] The TOF method (and several other analytical methods) can only screen for fluorine and not specifically for the source of fluorine (IF or OF).[37–39] Therefore, an isolation step is needed to remove IF, or to extract OF. Naturally, different analytical methods have different extraction procedures and thus isolate different amounts of IF or OF, meaning that the comparison between methods is difficult. Therefore, an analytical technique that is selective between IF and OF, and without an extraction step, can contribute to the TOF method to overcome its drawbacks. One of those techniques is fluorine-19 nuclear magnetic resonance (^{19}F NMR) spectroscopy.[38]

Fluorine-19 Solid-State Nuclear Magnetic Resonance Spectroscopy

In NMR spectroscopy, the magnetic properties of atomic nuclei are examined.[40] It is a powerful tool for the analysis of a wide range of systems and states of matter, including gases, liquids, gels, and solids. The most routinely used nuclei are ^{1}H and ^{13}C, but ^{19}F is also an excellent nucleus for NMR spectroscopy because of its high gyromagnetic ratio and 100% natural abundance, yielding very high sensitivity relative to other nuclei. Another significant advantage of using ^{19}F NMR spectroscopy is that it has a relatively large chemical shift dispersion, thereby allowing identification of the signals to a specific fluorine containing compound. Consequently, ^{19}F NMR is widely used both in solution state and solid-state NMR applications. Solid-state (ss-)NMR, as the name suggests, is the NMR technique for characterizing structures in solid matter (e.g., in powders, single crystals and amorphous materials).[41] The principles behind ss-NMR are similar to solution NMR; however, because of the orientation dependence of the nuclear spin interactions, samples are typically measured under magic-angle spinning (MAS), where the samples are spun at very fast speeds and at an angle of 54.7° with respect to the external magnetic field.[42,43] As a result, anisotropic interactions are averaged, thus improving the resolution and the sensitivity of the technique.[44] Solid state NMR is particularly valuable for samples that are not readily soluble in common NMR solvents, such as cellulosic and fiber-based substrates. Hence, ^{1}H and ^{13}C ss-NMR spectroscopy already have been applied in the analysis of paper and fiber-based applications.[45-49] Another important advantage is that ss-NMR eliminates the need for extractions and, therefore, does not rely on extraction efficiencies of analytes from solid matrices. The successful use of ^{19}F NMR analysis to identify specific PFAS in water treatment and environmental samples has been demonstrated, even when the substances were present only at parts per billion levels.[37,50–52] By using ^{19}F NMR, fluorinated components are differentiated by their resonances or chemical shifts due to their different fluorine environments. The presence of inorganic versus organic fluorinated substances is determined and then the substances are identified and classified accordingly, because PFAS have a unique fingerprint in the ^{19}F spectra for the -CF_2- and -CF_3 moieties.[37,52]

The solid-state approach was established in collaboration with the NMR staff at the University of Delaware. This method provides detection of fluorinated compounds in a solid-state

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cellulosic-based matrix and specifically for detection of PFAS. Cellulose does not contain any fluorine nuclei; therefore, no spectral overlap issues occur with the carbohydrate resonances, which simplifies the interpretation of organic versus inorganic fluorine containing components.

The goal of this study is to show the valuable insights that ^{19}F ss-NMR can provide and to demonstrate that it can complement different analyses, such as the TOF-method, particularly in cases of uncertainty. We conducted qualitative ^{19}F ss-NMR analysis on several raw materials and were able to confirm the presence of non-extractable inorganic fluorine, which would be interpreted as PFAS by the TOF method. We further demonstrated that PFAS levels below the indicator value of 20 mg/kg dw would be detected with ^{19}F ss-NMR analysis. This study provides a proof of concept for the practical application of ^{19}F ss-NMR spectroscopy in both class-specific and total analysis of fluorinated compounds in food packaging materials.

MATERIALS AND METHODS

Raw Materials, Coatings, and Paper Substrates

All coated and blank paper samples used in this study were supplied by our customer, who is a main supplier of wrapping paper for large fast-food restaurants across Europe. The blanco (untreated) paper was a white virgin fiber paper with a base weight of 25.5 g/m². The coated paper was produced with a 3.5 g/m² barrier coating on one side. The raw material talc was supplied by an industrial manufacturer. The barrier coating was Solenis' water based TopScreen™ DS 7G barrier coating, which followed strict quality control measures, and was manufactured at and supplied by the Lummen, Belgium production plant. The chemicals, perfluorooctanoic acid (PFOA, 95% purity, CAS 335-67-1), dimethyl sulfoxide-d_6 (d_6-DMSO, 99.9% atom %D, CAS 2206-27-1) and acetone-d_6 (99.9% atom %D, CAS 666-52-4), were purchased from Sigma-Aldrich.

Total Organic Fluorine Analysis

The TOF analyses were conducted by Eurofins (SAA-H-TOF.015:2018-11) and ISEGA (SOP 162.200) based on the European Standard for the characterization of halogen content by combustion (DIN EN 14582:2016-12). One gram of material was combusted in an oven at a temperature of 900°C under Argon flow. The HF was then absorbed into a 1:100 diluted eluent solution (Na_2CO_3 + $NaHCO_3$). Fluoride was determined by IC using 4-fluorobenzoic acid as a reference. To determine the TOF, inorganic fluorine was eliminated by water extraction.

Analysis of Inorganic Fluoride Fraction

The cold-water extraction was accomplished following the DIN EN 645:1994-01 extraction method. Then ion chromatography (IC) was performed following the ISO 10304-1 method. A sample of the barrier coating and the talc raw material were diluted at a ratio of 1:10 and filtered using a 0.45 µm filter. The coated paper samples were subjected to cold water for 24 hr and filtered with a 0.45 µm filter. All samples were prepared in triplicate to ensure reliability of results, and the average results and the deviation were reported. The diluted barrier coating product, the talc raw material, and the paper extract samples were then analyzed using an IC system (ThermoFisher Dionex ICS 5000) equipped with an anion exchange column (4 mm x 250 mm, Dionex AS 14-A) and a conductivity detector, using K_2CO_3 as eluent (Dionex ICS 5000 EG eluent generator). The flow rate was 1 ml/min at 23°C.

Sample Preparation for 19F Solid-State NMR Analysis

All samples were pulverized before they were submitted for solid-state NMR analysis. A 1 – 2 g sample, enough to provide a representative sample, was loaded into a SPEX 6750 cryogenic grinder. Paper samples were first cut into strips of a maximum of 5 cm long to ensure good grinding. The amount of material needed for the solid-state rotor depends on the size of the rotor. For 3.2-mm rotors, 50 mg of material is sufficient, while for 4-mm rotors, approximately 100 mg is needed. For 1.3-mm rotors, less than 5 mg of material is necessary. The cryo-milled papers yielded homogeneous powders that were packed into various sized NMR rotors using a funnel and standard rotor packing tools from Bruker.

Sample Preparation of PFOA Spiking

A 0.2018 g sample of a coated paper (lot X-00256-23-1) from the cryo-mill grinder was weighted into a 5 ml vial. For the spiking solution, 0.0604 g PFOA was dissolved in 2.4671 g of DMSO. The percentage of actives (PFOA) in the initial solution was 2.39. A second solution was prepared by diluting 0.0222 g of the first solution in 2.491 g of DMSO. The percentage of actives (PFOA) in the second solution was 0.0211. From the second solution, 0.0236 g was weighed and dissolved in 1 g of acetone. Thus, the final solution contained 5 µg of PFOA. Perfluorooctanoic acid has a molecular weight of 414 g/mol, and the total molecular weight of fluorine is 285 g/mol; therefore, elemental fluorine represents 68.8% wt of the molecular weight. Thus, the total weight of elemental fluorine in the final solution was 3.43 µg. The final solution was added to the 5 ml vial containing the 0.2018 g of cryo-milled coated paper. Acetone was left to evaporate, and then the entire sample was submitted for ^{19}F NMR analysis. Thus, the peaks correlating to PFOA in the ^{19}F spectra translated to an elemental fluorine content of 17 mg/kg.

Fluorine-19 Solid-State NMR Analysis

Solid-state ^{19}F magic angle spinning (MAS) ss-NMR spectra were acquired on a 11.7 T (500 MHz) wide bore Bruker Avance III spectrometer with a 4-mm H/FX probe at a spinning rate of 15 kHz. Typical ^{19}F 90° pulse lengths were 2.7 µs. Although no formal T_1 measurements were performed given the goal of qualitative identification of ^{19}F species (in addition to the very low sensitivity particularly for PFAS), multiple relaxation delays were tested. A delay of 5 seconds found was to be a reasonable compromise between sensitivity and experiment time and is sufficient for detection of both PFAS and talc peaks. (See Figure S1, Supporting Information.) No ^{1}H decoupling was necessary. The number of transients was between 512 and 14336 scans. The experiment time ranged between 43 minutes and 20 hours. The acquisition time was approximately 15 ms. During the odd-numbered transients, a 180° pulse was applied before the 90° excitation pulse (180-τ-90; φ_{180}: 0; φ_{90}: 0, 0, 1, 1, 2, 2, 3, 3; $\varphi_{\text{-receiver}}$: 0, 2, 1, 3, 2, 0, 3, 1) to suppress the ^{19}F background signals from the 4 mm probe and the Kel-F rotor caps. (See Figure S2, Supporting Information.) An inter-pulse delay (τ) of 10 ms was used. This pulse sequence and delay resulted in an approximately 10% loss of sensitivity relative to a Bloch decay experiment (90-acquire; see Figure S3, Supporting Information). Non-fluorine containing Vespel rotor caps were used when possible. Spectral widths in these experiments were 300 ppm (140 kHz at 500 MHz, 170 kHz at 600 MHz) using ^{19}F excitation pulse powers of ~90 kHz (typical for ss-NMR), yielding maximum resonance offsets at the edges of the spectra of ~70-85 kHz. In our case, the goal was qualitative rather than quantitative detection of ^{19}F and thus the excitation bandwidth of these pulses was sufficient. Spectra at 14.1 T (600 MHz) were acquired on a standard bore Bruker

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Avance III spectrometer with a 3.2-mm H/FX probe at a spinning rate of 24 kHz both with and without the baseline suppression pulse sequence described previously. This probe did not displayed the ^{19}F background signals that were observed with the 4-mm probe (see Figure S2, Supporting Information); thus, the baseline suppression sequence was not necessary, and a simple 90-acquire sequence was used. Typical ^{19}F pulse lengths were 2.8 µs. Spectra at 60 kHz MAS were acquired on a 11.7 T (500 MHz) wide bore Bruker Avance III spectrometer with a 1.3mm HFX probe using the standard Bloch 90° pulse-acquire sequence. Typical ^{19}F pulse lengths were 2.3 µs. During acquisition 15 kHz two-pulse phase-modulation (TPPM) ^{1}H decoupling was applied.

The raw NMR data was processed using Bruker Topspin software. Apodization, or line broadening, of a 100-Hz was applied to reduce noise and make any relatively weaker resonances easier to observe.

RESULTS AND DISCUSSION

Total Organic Fluorine Analysis

Eighteen TOF analyses were conducted on different lots of coated and uncoated papers that were produced between 2020 and 2021. All coated papers were coated with the barrier coating; measurements were taken at different points in time and at different consumer test service laboratories. Figure 1 shows that the TOF values of the uncoated papers were less than the indicator value of 20 mg/kg dw TOF, and the values of all the coated papers were greater than the indicator value.

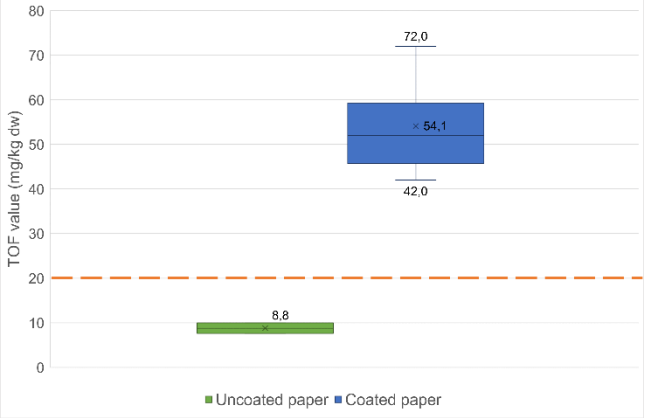


Figure 1. The TOF values from coated and uncoated paper samples compared with the Danish indicator value, shown by the orange dotted line.

This meant that the coating was responsible for the higher TOF values and that it potentially contained PFAS because the TOF method is considered to be an accepted proxy for the total PFAS content for fiber-based substrates.[17] However, no compounds that contained PFAS were directly added to the water-based coating to enhance its barrier properties. One possibility was that PFAS were indirectly added to the coating or paper during manufacturing, through lubricants of coating equipment, additives in raw materials, and so forth. Another possibility was that the TOF method measured more than just organofluorines because the extraction step did not remove all inorganic fluorides before the combustion step and that caused an overestimation of the TOF values.

The barrier coating in question consists of three major components, a polymeric binder, a talc and a wax, and minor components such as additives. Talc is a conventional inorganic filler used in barrier coatings to reinforce the coating, enhance the barrier properties, or lower the cost of the overall formula. Talc is a hydrated magnesium silicate, a trioctahedral layered mineral, with the chemical formula $Mg_3Si_4O_{10}(OH)_2$. Within the talc crystal structure several other elements are found in smaller quantities or traces, such as naturally occurring fluoride ions (F^-) substituting hydroxyl groups (OH^-) [53,54].

Table 1. Total fluorine determined on different talc batches.

Talc Batches	Total Fluorine Measurements (mg/kg dw)			
	A	B	C	D
895249	710	750	780	810
895250	790	800	830	800
895251	450	440	410	390
895255	580	660	/	/
895256	640	710	/	/
895264	760	750	/	/
895265	510	510	/	/
Total average:	**654 ± 151 mg/kg dw**			

To examine the amount of fluoride that can be present in talc, TOF measurements were performed on the raw material, without the water extraction step. Seven different batches of talc were sampled. The first three batch samples were measured four times, while the other four batches were only measured two times. All TF values are given in Table 1. The TF contents of the talc raw material were higher than expected because the supplier's data sheets reported that the IF content in the talc didn't exceed 400 mg/kg. However, these values are consistent with the TOF values of the coated paper when back calculated from coated paper to the coating (~12% wt of the coated paper) and to raw material talc (~55% wt of the coating), without the subtraction of the IF content.

Analysis of Inorganic Fluorides

To validate the extraction step of the TOF method, a cold-water extraction followed by IC analysis was performed on the coated papers, the water-based coating itself and the raw material talc. It is expected that the IF extraction corresponds to the TF content, because it is assumed that the talc does not contain any other form of fluorine except for the inorganic F^-. Four different batches of the papers, coatings and talc were tested (resulting in 12 data points per sample category). Figure 2 shows the results. The variance increased from the talc data set to the coating data set and from the coating data set to the coated paper data set. A big difference in the results occurs between the IF values from the cold-water extract and the TF values determined through CIC.

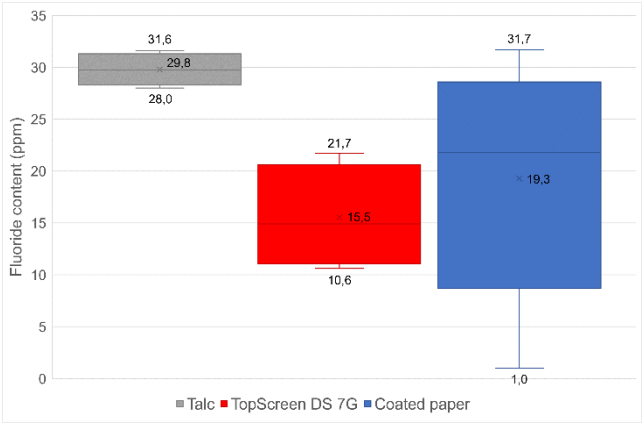


Figure 2. The inorganic fluoride concentrations in mg/kg of different sample categories by cold-water extraction followed by ion chromatography analysis.

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This could mean that the talc raw material does contain other forms of fluorine, such as PFAS from drilling fluids for example. However, the difference between the IF and TF values could also mean that the water extraction step is not sufficient to extract all the fluoride embedded in the crystal structure. In addition, the analysis showed a correlation between the fluoride content in the pure talc samples and the coating samples that was attributed to the amount of talc in the coatings. Inconsistencies in the fluoride levels of the cold-water extract from the coated paper could suggest the presence of other sources of fluoride in the paper in addition to the coating layer.

Also, factors such as the inherent heterogeneity of paper and the manufacturing process could have contributed to these variations because the talc was finely dispersed into the coating particles and afterwards treated and cured as a coating layer onto the paper. Therefore, correlation does not imply causation and the low accuracy and precision of the extraction step of the coated paper samples made it difficult to draw conclusions. To resolve the inconsistencies and more accurately distinguish OF and IF fractions, ^{19}F ss-NMR was pursued.

Fluorine-19 Solid-State NMR Analysis

Figure 3 shows different ^{19}F ss-NMR spectra of known and unknown samples relevant to the study. While the general chemical shift range for ^{19}F NMR is approx. 700 ppm, PFAS typically appear over a 250-ppm region (+50 to -200 ppm). The purple spectrum in Figure 3 was that of calcium fluoride. The chemical shift of the fluorine ions in CaF_2 occurred at -108 ppm, which was in excellent agreement with the literature reference of CaF_2.[55] This compound was suspected to be present in the analyzed samples of paper; therefore, the spectrum was obtained in anticipation of finding this signal. Note that spinning side bands occurred at a distance that was based on the spinning speed of the solid-state rotor.[56] In this case, the spinning speed of the rotor was 15 kHz, and the data was acquired at an operating frequency of 470.5 MHz. Hence, the first order spinning side bands occurred at ± 31.9 ppm and were visible on each side of the central signal annotated by the asterisk. The ^{19}F spectrum of a blank, untreated, paper sample from the commercial paper supplier (see the green spectrum in Figure 3) showed no peaks throughout the entire chemical shift range. This was expected because the compounds that make up untreated paper (cellulose fibers, starches, sizing agents, etc.) do not contain any form of fluorine. A paper sample from the commercial paper supplier known to contain PFAS served as an excellent proxy to check if solid-state analysis by ^{19}F NMR would be a viable approach for the detection of fluorinated compounds in paper. The ^{19}F spectrum is provided as the orange spectrum in Figure 3. Literature references state that the $(-CF_2-)_n$ repeat moiety on a perfluoroalkyl chain typically is observed between -115 and -125 ppm. The signal that corresponds to the terminal trifluoromethyl, CF_3-R, typically is observed between -80 and -85 ppm in the ^{19}F spectrum. The ^{19}F spectrum displays a chemical shift pattern that agrees with the cited literature references.[37] Furthermore, the intensity ratio of $-CF_3$ versus $(-CF_2-)_n$ moieties provides a qualitative estimate of the alkyl chain length on a relative basis. Note the presence of several sets of spinning side bands (*), ± 31.9 ppm from their central signals. Samples of coated hamburger wrapping paper used for the TOF tests were also submitted for ^{19}F ss-NMR analysis. One of the obtained spectra is shown in Figure 3 as the blue spectrum. A single peak is observed at a chemical shift that occurred at -175 ppm. This shift was from a completely different fluorine component than PFAS or calcium fluoride.

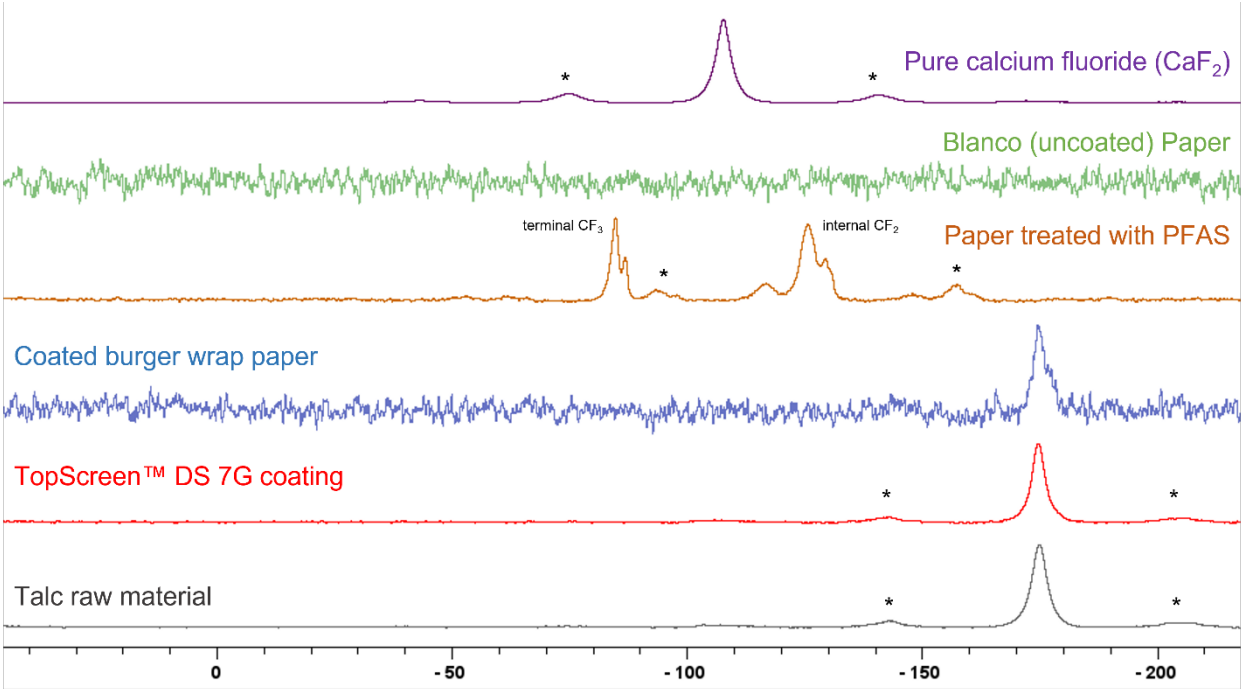


Figure 3. Fluorine-19 solid-state NMR spectra of different substances. Purple: pure calcium fluoride (CaF_2), clear signal at -108 ppm. Spinning side bands are indicated with an asterisk. Green: spectrum of uncoated and untreated base paper from commercial supplier, no signal. Orange: paper from commercial supplier treated with PFAS compound(s), signals between -80 and -85 ppm represent the $-CF_3$ moieties and signals between -115 and -125 ppm represent the $-CF_2-$ moieties. Blue: coated paper from commercial supplier, coated with TopScreen DS 7G barrier coating, signal at -175 ppm but high signal-to-noise ratio (SNR). Red: sample dried out water-based coating TopScreen DS 7G barrier coating, clear signal at -175 ppm. Grey: talc raw material, clear signal at -175 ppm. All spectra were acquired at 11.7T, 15 kHz MAS with a 4 mm rotor using the baseline suppression pulse sequence. The number of transients for all spectra was 512 scans (experiment time of 43 minutes).

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The blanco paper (green spectrum) exhibited no ^{19}F chemical shifts; therefore, the fluorine signal in the coated wrapping paper was derived from the coating. Note that the signal-to-noise ratio (SNR) was quite low for the blue spectrum, indicating a relatively low overall amount of the fluorine components. The red spectrum in Figure 3 shows the ^{19}F chemical shifts of the pure barrier coating. The coating was cured and dehydrated for 72 hours in an oven at 120°C before the residue was cryo-milled into a powder. The exact same signal that appeared at -175 ppm in the coated paper also appeared in the pure coating. This provided direct evidence that the fluorine-containing component in the coated papers originated from the coating. The spinning side bands were revealed because of the higher intensity of the signal. All the raw components in the barrier coating were evaluated; only the talc raw material showed a signal in the ^{19}F spectrum (grey spectrum), and it was the exact same signal at -175 ppm (with the same spinning side bands) as in the coated paper sample and the pure coating sample. This ^{19}F signal of -175 ppm can be assigned to fluoride atoms in the structure of talc.[57,58] Chemical shifts of relatively more positive values indicate downfield shifts, lower-shielding, or higher frequency; chemical shifts of relatively more negative values indicate upfield shifts, higher-shielding, or lower frequency. In the talc matrix, the F^- ions is shielded by the {Mg-Mg-Mg}-sites in the crystal structure, hence an upfield shift occurred compared with the signals of PFAS (wherein the $-CF_3$ moieties are less shielded than the $(-CF_2-)_n$ moieties) or the signals of CaF_2 (which has a less dense octahedral crystal structure). Hence, ss-NMR indicates that what was interpreted to be organic fluorine by the TOF method actually is non-extractable inorganic fluorine. To ensure the absence of PFAS and OF in these food contact papers, a PFAS-doping study was established to confirm that levels greater than the indicator value of 20 mg/kg dw would be detected in these ss-NMR experiments.

The Effect of Spiking on 19F Solid-State NMR Spectroscopy for the Detection of PFAS

One of the drawbacks of using ^{19}F NMR spectroscopy is the low limit of detection (LOD) that is inherent with this technique. Because of anisotropic interactions and low resolution during solid-state measurements, magic angle spinning is crucial to produce more narrow and intense NMR shifts. On the other hand, an advantage of the detection of PFAS is that the $(-CF_2-)_n$ moiety is observed typically between -115 and -125 ppm and $-CF_3$ between -80 and -85 ppm in the ^{19}F spectrum. Even if a sample contained a complex mixture of different PFAS, the chemical shifts of the $(-CF_2-)_n$ and $-CF_3$ moieties would result in a broad complex signal but with enough intensity to be qualitatively detected above the noise (vide infra).

To illustrate the capability of ^{19}F ss-NMR spectroscopy to detect low levels of PFAS, a PFOA-spiked coated paper sample was analyzed for comparison. A quantity of 17 mg/kg of elemental fluorine (via PFOA) was added to the paper sample, which was less than the Danish indicator value of 20 mg/kg dw TOF to prove that ^{19}F ss-NMR could at least distinguish between intentionally and unintentionally added PFAS. The NMR spectra of the clean coated paper and the spiked sample with PFOA are illustrated in Figure 4. The number of transients for the coated hamburger wrapping paper in Figure 3 was 512 scans; however, in Figure 4 the number of transients was increased to 14336 scans to improve the SNR. The ^{19}F signal patterns for the $(-CF_2-)_n$ and $-CF_3$ moieties were observed at their respective chemical shifts in the spiked sample, while they were totally absent in the control sample.

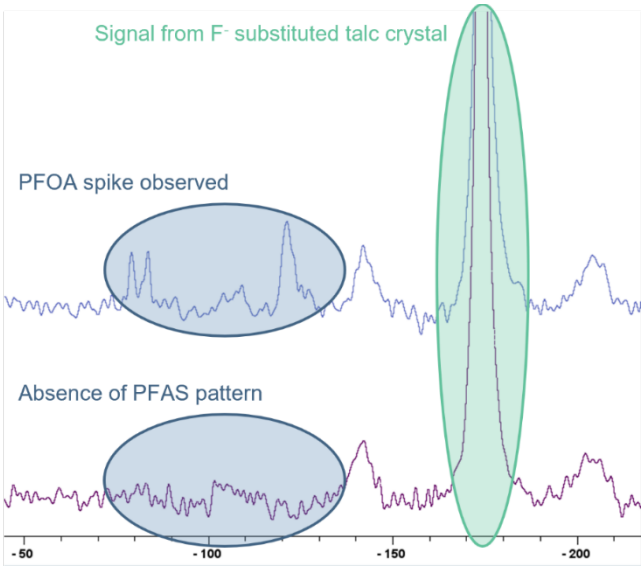


Figure 4. Comparison of ^{19}F spectra between spiked and control coated paper samples. Blue spectrum: coated paper sample spiked with 17 mg/kg elemental fluorine (via PFOA). Purple spectrum: PFAS-free control coated paper sample. Both spectra were acquired at 11.7 T, 15 kHz MAS with a 4-mm rotor using the baseline suppression pulse sequence with 14336 scans (experiment time of 20 hours).

Thus, ^{19}F ss-NMR provides a class-specific determination as direct spectroscopic evidence for PFAS and inorganic fluorinated compounds. Note that the baseline perturbation between -100 and -110 ppm in both spectra was attributed to the second order spinning side bands of the main signal at -175 ppm.

Improving Spectral Quality and Increasing Sensitivity

Further characterization was performed to determine the optimal set of acquisition parameters for future work with a particular focus on the amount of material packed into the NMR rotor and hardware selection, such as probes for faster MAS rates. Generally, because of the relatively large ^{19}F chemical shift anisotropy (CSA), faster MAS is preferred for more efficient averaging of the CSA and hence improved sensitivity. However, in the case of very dilute species, a larger sample amount is preferred (i.e., larger rotors).

A 3.2-mm rotor holds approximately 40 – 50 mg of material and can spin up to 24 kHz. A 4-mm rotor holds 100 mg of material but is limited to a 15 kHz maximum MAS rate. A 1.3-mm diameter rotor holds approximately 2 – 5 mg of material and can spin to more than 60 kHz, The faster spinning speed accessible with a 3.2-mm rotor, relative to a 4-mm rotor (24 vs 15 kHz), yields improvement in sensitivity, even with a 50% reduction in the amount of material packed into the rotor. However, moving to smaller rotor diameters and faster spinning yield diminishing returns. Overall sensitivity is much lower for 5 mg of a sample packed into a 1.3-mm rotor spinning at 60 kHz MAS, even with additional gains through the application of ^1H decoupling. For this application of fluorinated compounds in fiber-based substrates, the optimal sensitivity was obtained with a 3.2-mm rotor spinning at 24 kHz, even when correcting for the higher B_0 field strength (a factor of approximately 1.3 sensitivity gain between 11.7 and 14.1 T), as demonstrated for the ^{19}F signal of 17 mg/kg PFOA-spiked paper in Figure 5. All spectra were acquired with 2048 scans and a 5 s pulse delay. The spectrum at 15 kHz used the baseline suppression sequence and the spectrum at 24 kHz was acquired with the standard

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Bloch 90° one pulse-acquire sequence with 15 kHz TPPM ^{1}H decoupling during acquisition. Additional features within the inorganic fluorine peak were observed with fast MAS (60 kHz) and 1H decoupling that were not present at slower MAS rates. A higher sensitivity (14336 scans, 20 hours experiment time) comparison of 15 versus 24 kHz MAS is shown in the Supporting Information Figure S4.

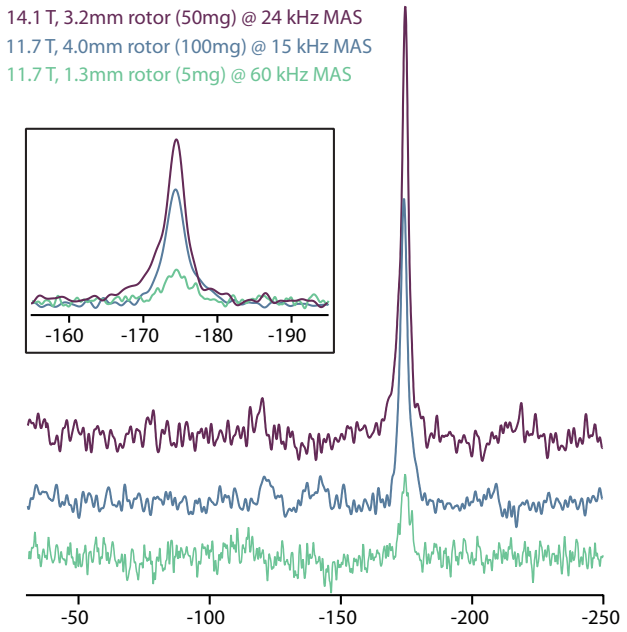


Figure 5. Comparison of sample mass, MAS rate, and field strength dependence of ^{19}F signals for the 17 mg/kg PFOA-spiked paper sample. Purple: 50 mg packed into a 3.2-mm rotor; signal-to-noise ratio (SNR) of 30, SNR/mg of 0.6. Blue: 100 mg packed into a 4-mm rotor; SNR of 25, SNR/mg of 0.25. Green: 5 mg packed into a 1.3-mm rotor; SNR of 5, SNR/mg of 1. Inset: overlay of the three spectra without offset, expanded to the fluorinated humite peak, indicating the gain in sensitivity at 14.1 T and 24 kHz MAS. All spectra were acquired with 2048 scans and 5s relaxation delay (experiment time of 3 hours).

While the goal of this work was to demonstrate that the qualitative detection of low concentrations of PFAS in food contact materials is possible with ss-NMR, the application can readily be extended to quantification of PFAS and other ^{19}F species with careful optimization of experimental parameters including pulse lengths and the relaxation delay (such that the relaxation delay is equal to 5 times the longest T_1 in the sample). For nuclei with a large chemical shift dispersion such as ^{19}F, the excitation bandwidth of the pulses also needs to be considered. Canonically, pulse offset effects become a concern when the resonance offset is on the order of the pulse RF power. High power excitation pulses are typically used in ss-NMR; therefore, the accessible bandwidth for a single experiment is still relatively broad. Offset effects compound when baseline suppression is required, because resonance offset occurs with both the 180° and 90° pulses. Supporting Information Figure S5 demonstrates excitation pulse offset effects and bandwidth limitations. Using the baseline suppression sequence, ~10% of signal is lost with an offset of 27 kHz, while in the extreme case of a 90 kHz offset and 90 kHz excitation pulse, an almost 50% reduction in signal is observed. For accurate quantification, spectra must have a sufficiently high SNR. Because of the very long experiment times that would be required to meet all these criteria, quantification of the PFAS signals was not pursued.

CONCLUSIONS

In this work, ^{19}F solid-state NMR was evaluated for the analysis and detection of fluorinated compounds in fiber-based consumer and food packaging samples, as a correction on the underestimation and overestimation of TOF values for total PFAS analysis. The characteristic chemical shifts for each tested sample were determined and were repeatable throughout different samples and their raw materials. Furthermore, ^{19}F ss-NMR spectroscopy was proven to have the capability to detect PFAS in quantities less than the indicator value of 20 mg/kg dw TOF and to differentiate between 'intentional' and 'unintentional' added PFAS. The sample matrix had a minimal impact on the signal intensity or chemical shifts of ^{19}F, thereby providing an important opportunity for detecting PFAS in more complex samples without the extensive clean up that typically results in sample losses when using other highly sensitive spectroscopic methods. Finally, the spectral quality and sensitivity were improved by modifying rotor and spinning speeds, resulting in lower acquisition times and lower limits of detection. Hence, solid-state NMR not only provides a direct observation for PFAS moieties but also fills an important gap in the analytic toolbox for food contact materials.

Future research should evaluate further optimization of instrument parameters and equipment and the use of ^{19}F internal standards to improve quantitative NMR assessment of total and class-specific fluorinated compounds in consumer and food packaging samples.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website.
Additional information about suppression of NMR background signals, signal loss, baseline pulse sequences and the influence of different NMR parameters on the signal strengths (pdf).

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ACKNOWLEDGMENT

The authors thank Solenis LLC for funding this project, and we acknowledge the University of Delaware for their contributions and measurements of ^{19}F solid state NMR.

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